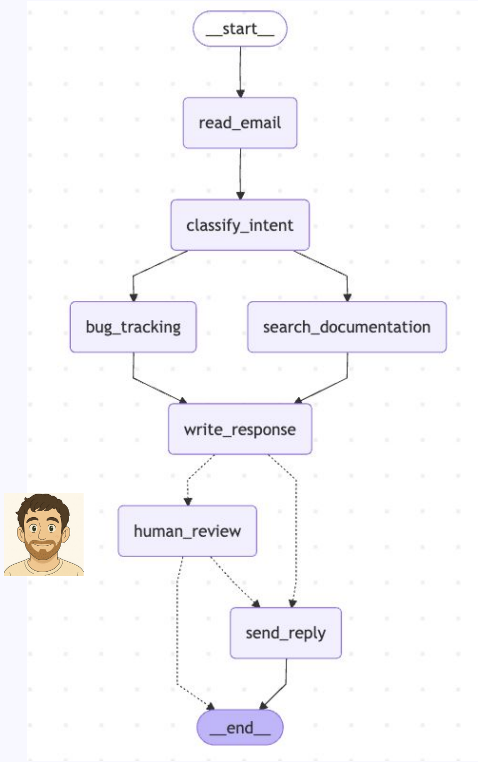
LangChain

LangGraph: StateGraph Essentials

Course Outline

1. LangGraph Orientation
2. LangGraph foundations
3. Build an Application



LangGraph Orientation

LangGraph

Agents and LLM applications have these challenges

- Latency in the seconds vs ms
  - Parallelization – to save actual latency
  - Streaming – to save perceived latency

https://blog.langchain.com/building-langgraph

Layer Diagram

Your Application

StateGraph SDK

Functional SDK

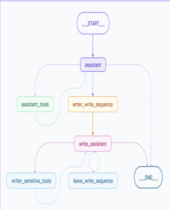
PregelLoop - Message Passing Runtime

LangGraph

LangChain

Hosted on LangSmith Deployments

Debugger



LangSmith

- Debugging
- Playground
- Prompt Management
- Annotation
- Testing
- Monitoring

COMMERCIAL

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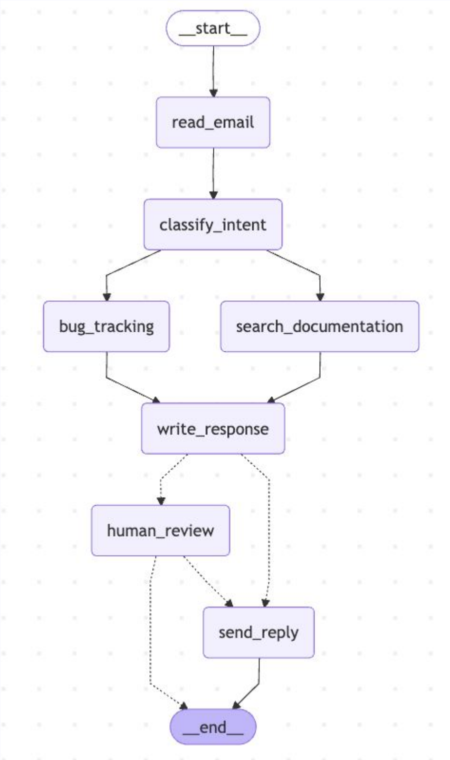


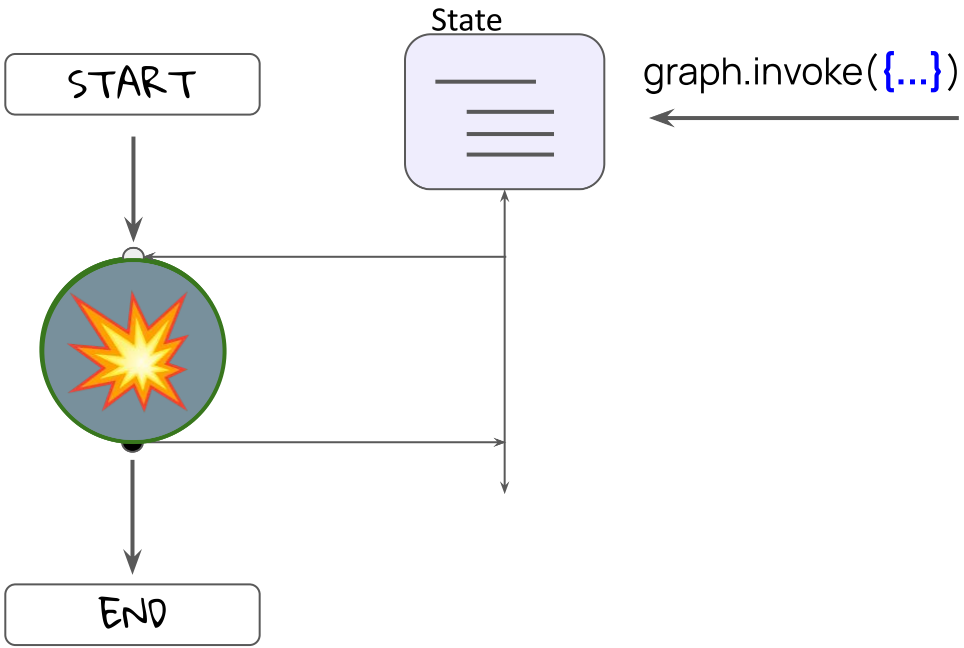
LangGraph: StateGraph

Components and Capabilities

- State: Data
- Node: Functions
- Edges: Control Flow
  - Serial, Parallel
  - Conditional
- Checkpointing/Memory
- Human In the Loop: Interrupts

Nodes and State

State, Nodes




```
const State = z.object({
  nlist: z.array(z.string())
});
```

```
function nodeA( state: State ) {
    ...
    return { nlist: [note] }
}
```

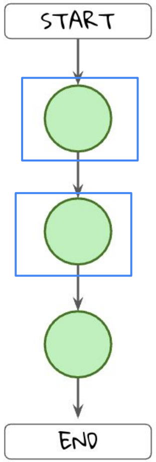
Edges

Edges: Control Flow

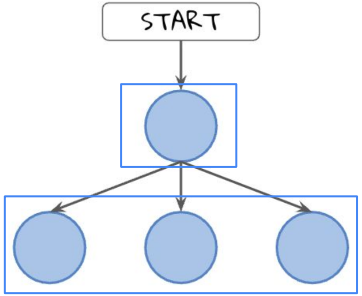
Edge

Serial

superstep

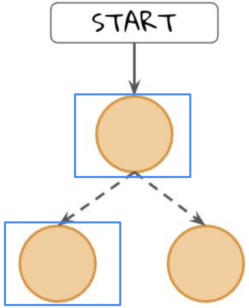


Parallel

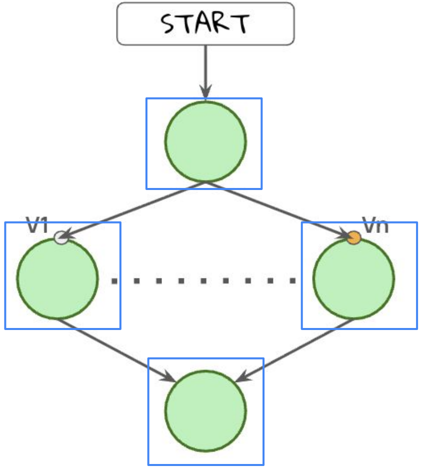


Conditional Edge

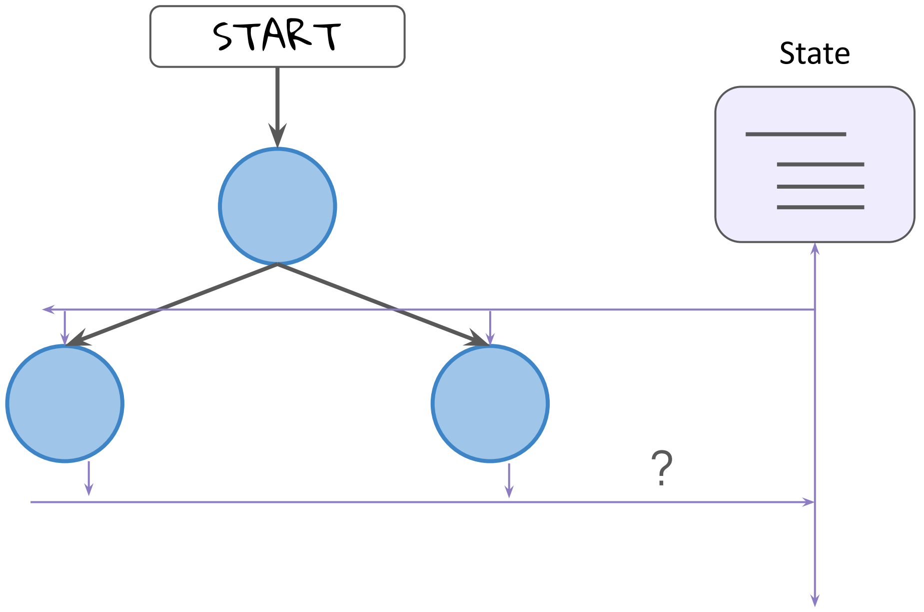
Conditional



Map-Reduce

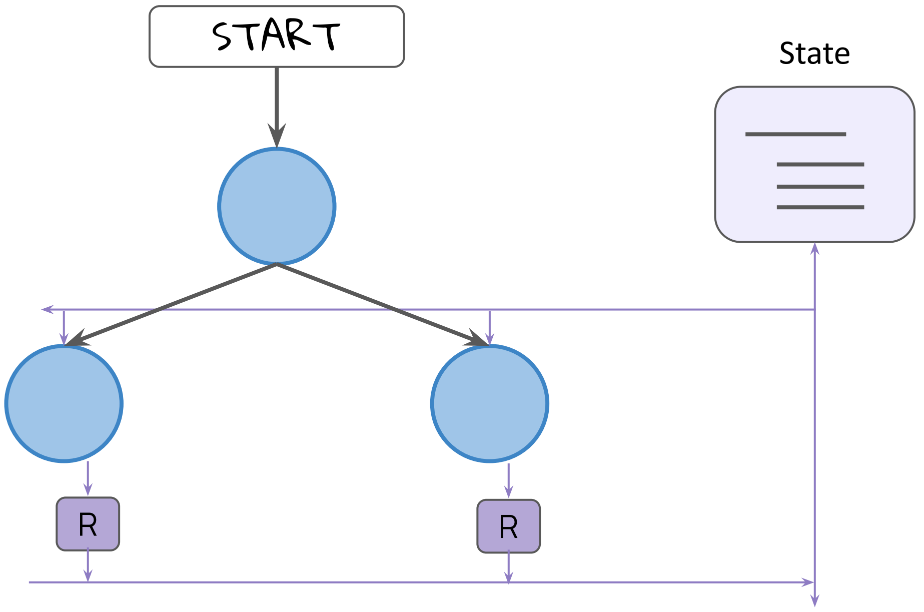


Reducers




```
z.object({
  nlist: z.array(z.string())
});
```

Reducers

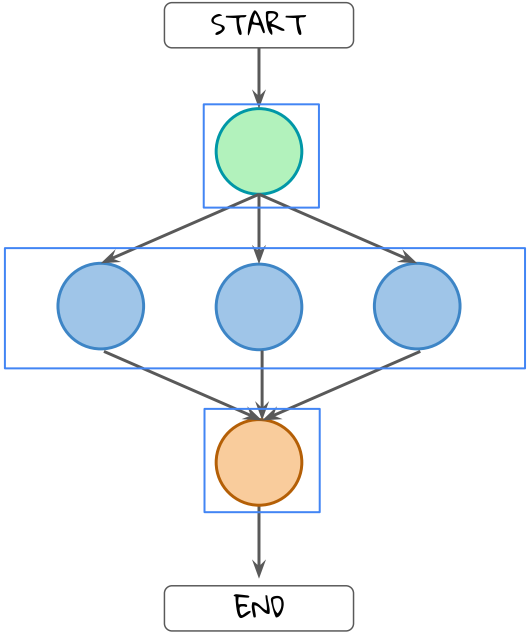


type

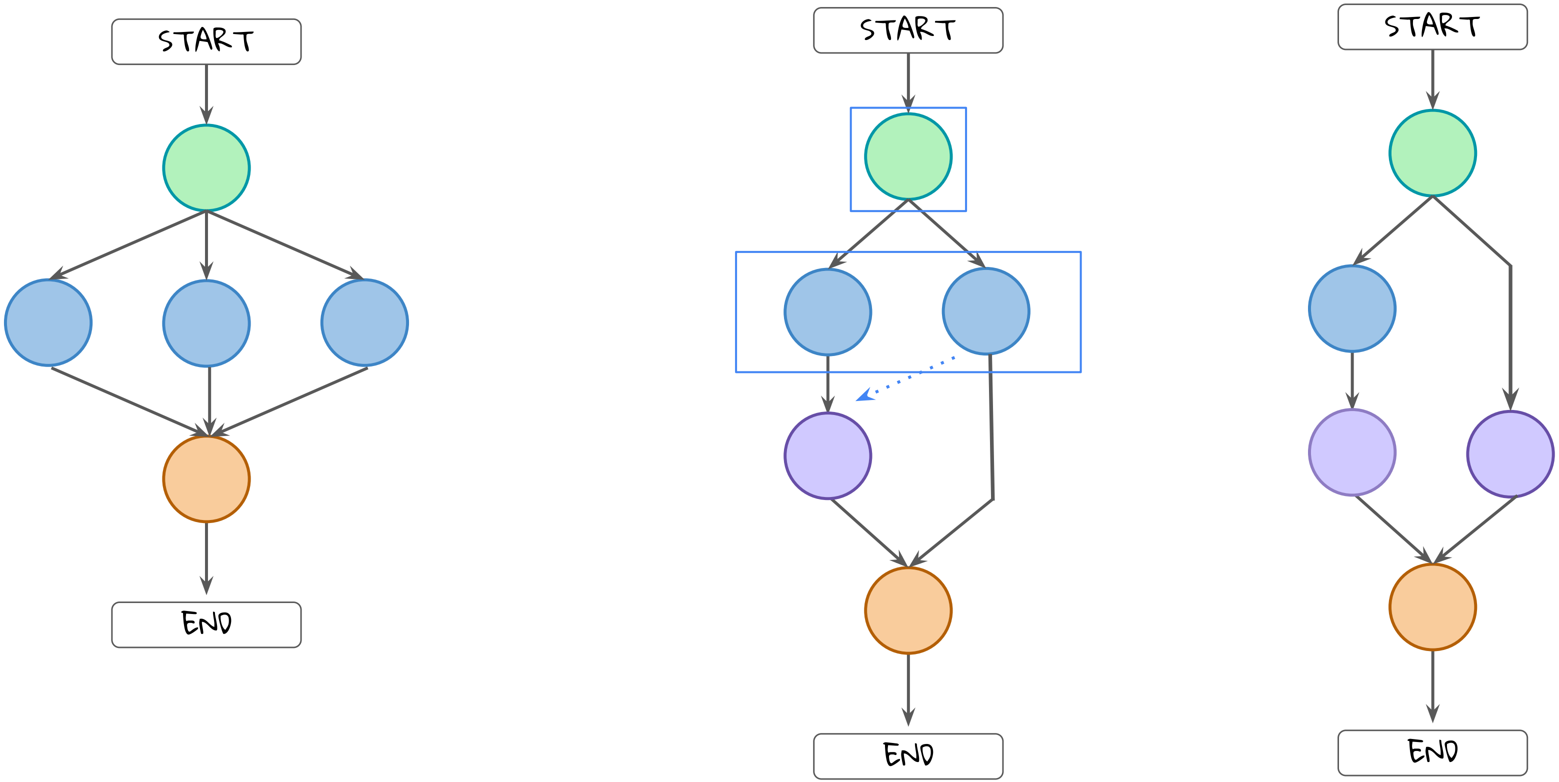
```
z.object({
  nlist: z.array(z.string())
    .register(registry, {
      reducer: {
        fn: (a, b) => a.concat(b)
      }
    }),
});
```

reducer function

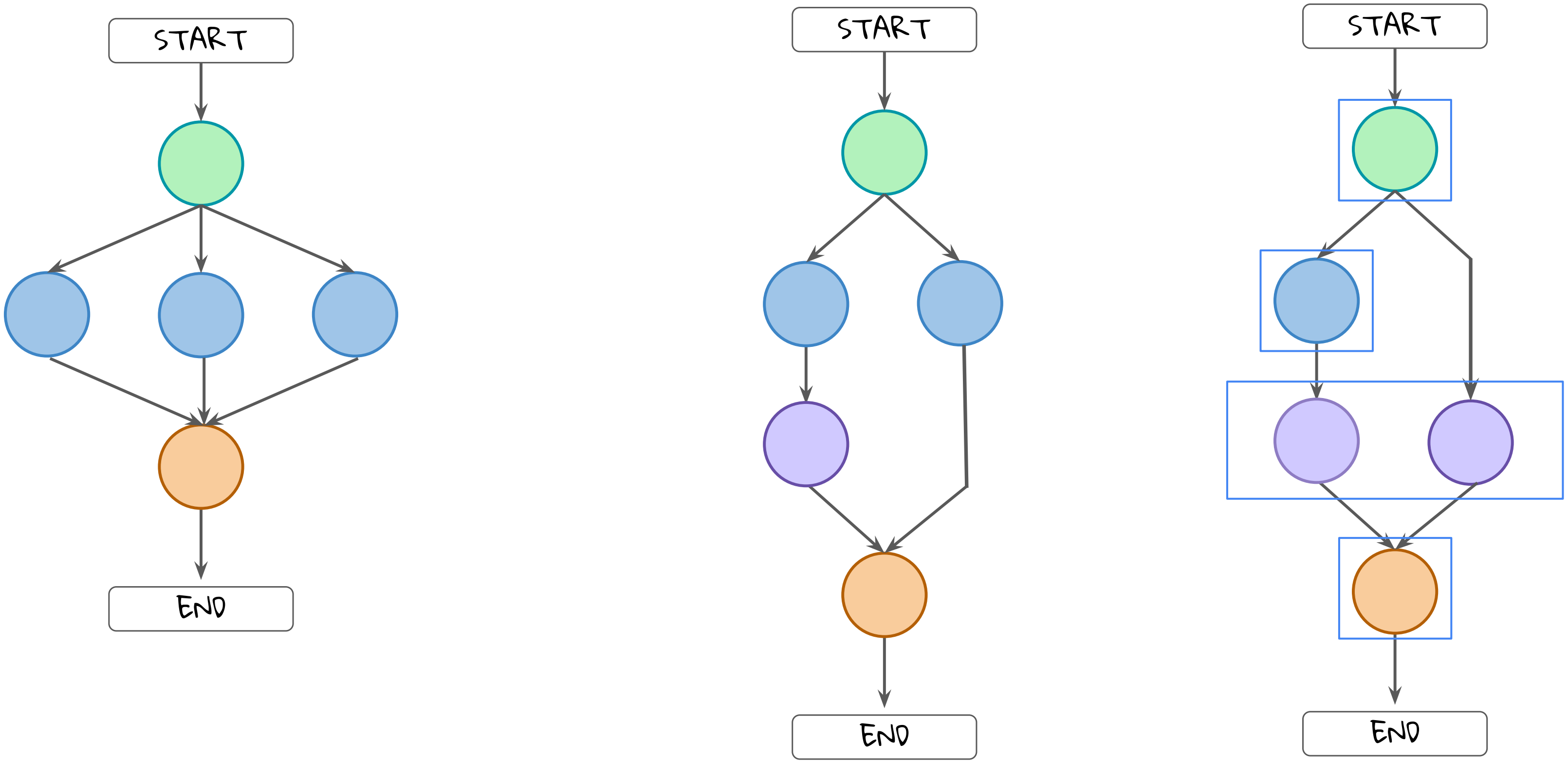
Super Steps



Super Steps



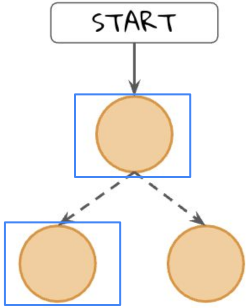
Super Steps



Conditional Edges

Edges: Control Flow

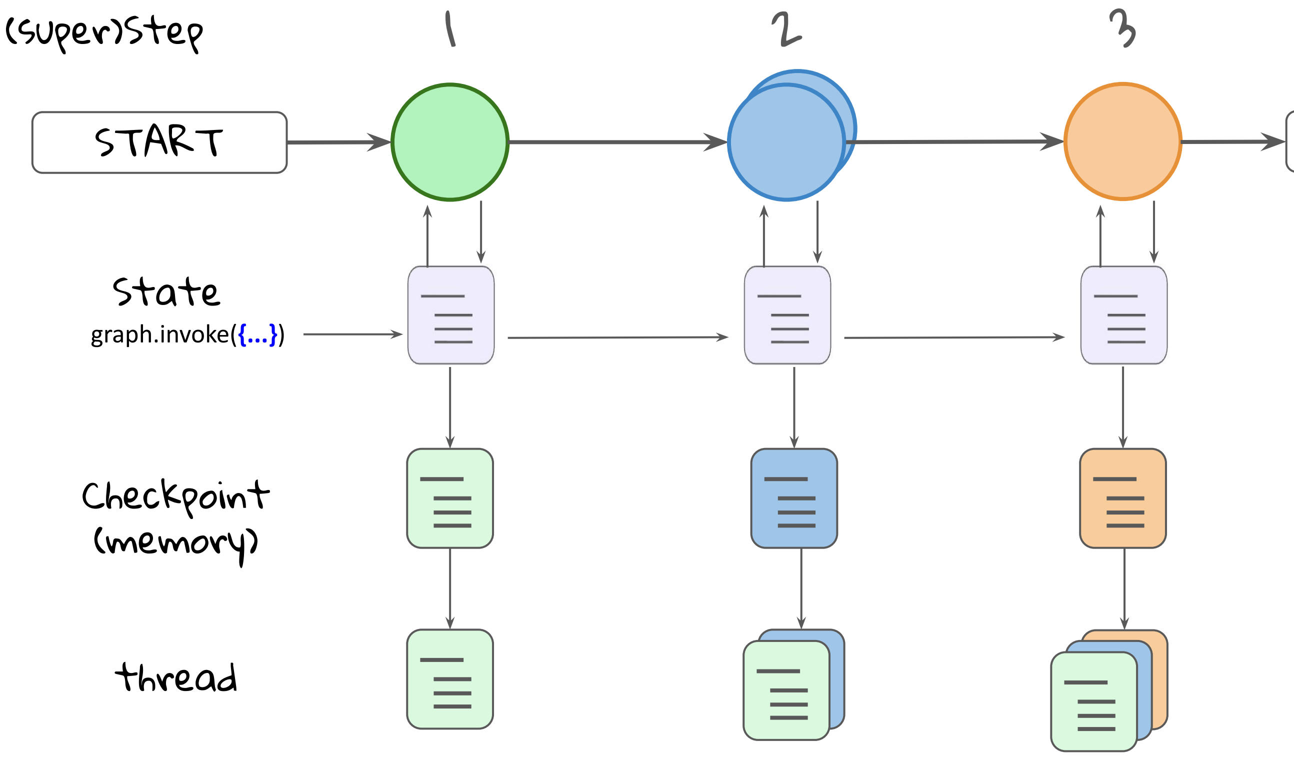
Conditional



- Conditional Edge
- Comman

Memory

Memory / Checkpointers



Nodes execute in sequence, per step

State is provided to the node at the start of a step and updated at the end of a step

State is stored to a **checkpoint** at the end of a step

A **thread** identifies a collection of checkpoints

Memory

Benefits

- **Recover gracefully from failures** — resume without losing progress.
- **Time travel** — roll back to a known good point and continue forward.

Memory / Checkpointers

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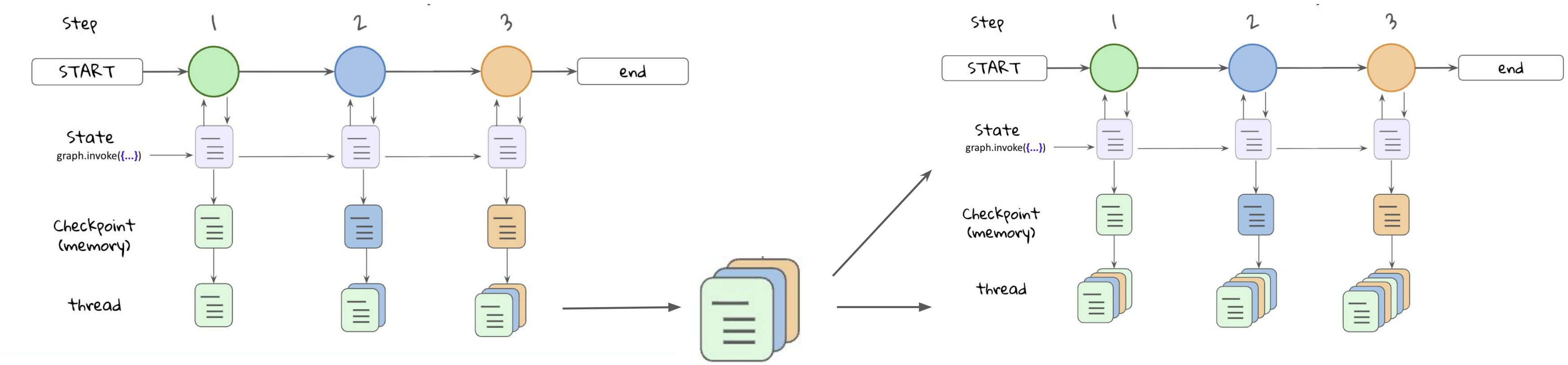
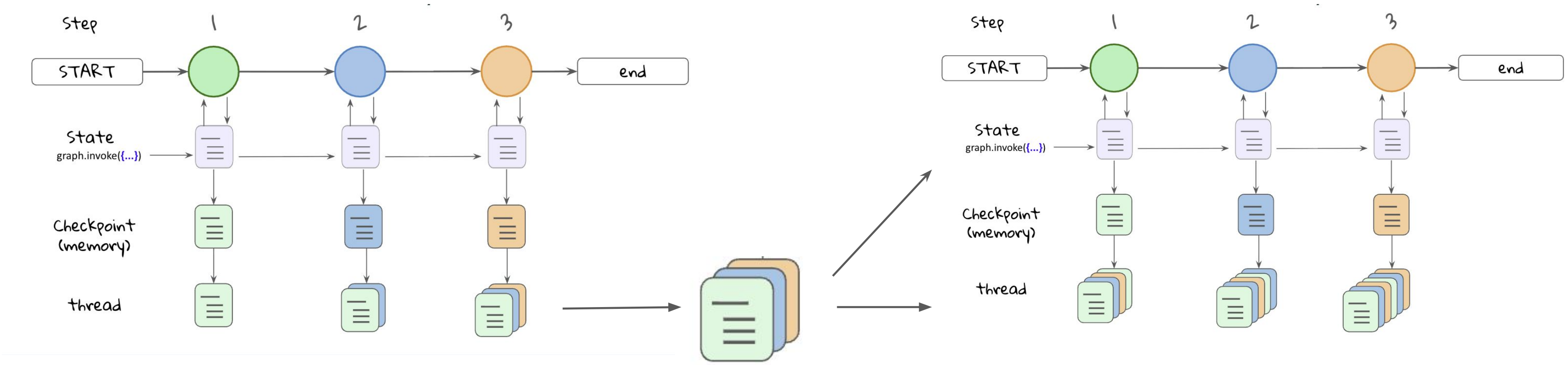
- **Recover gracefully from failures** — resume without losing progress.
- **Time travel** — roll back to a known good point and continue forward.
- **Persistent state** — data is preserved even when the graph is not running.



Memory / Checkpointers

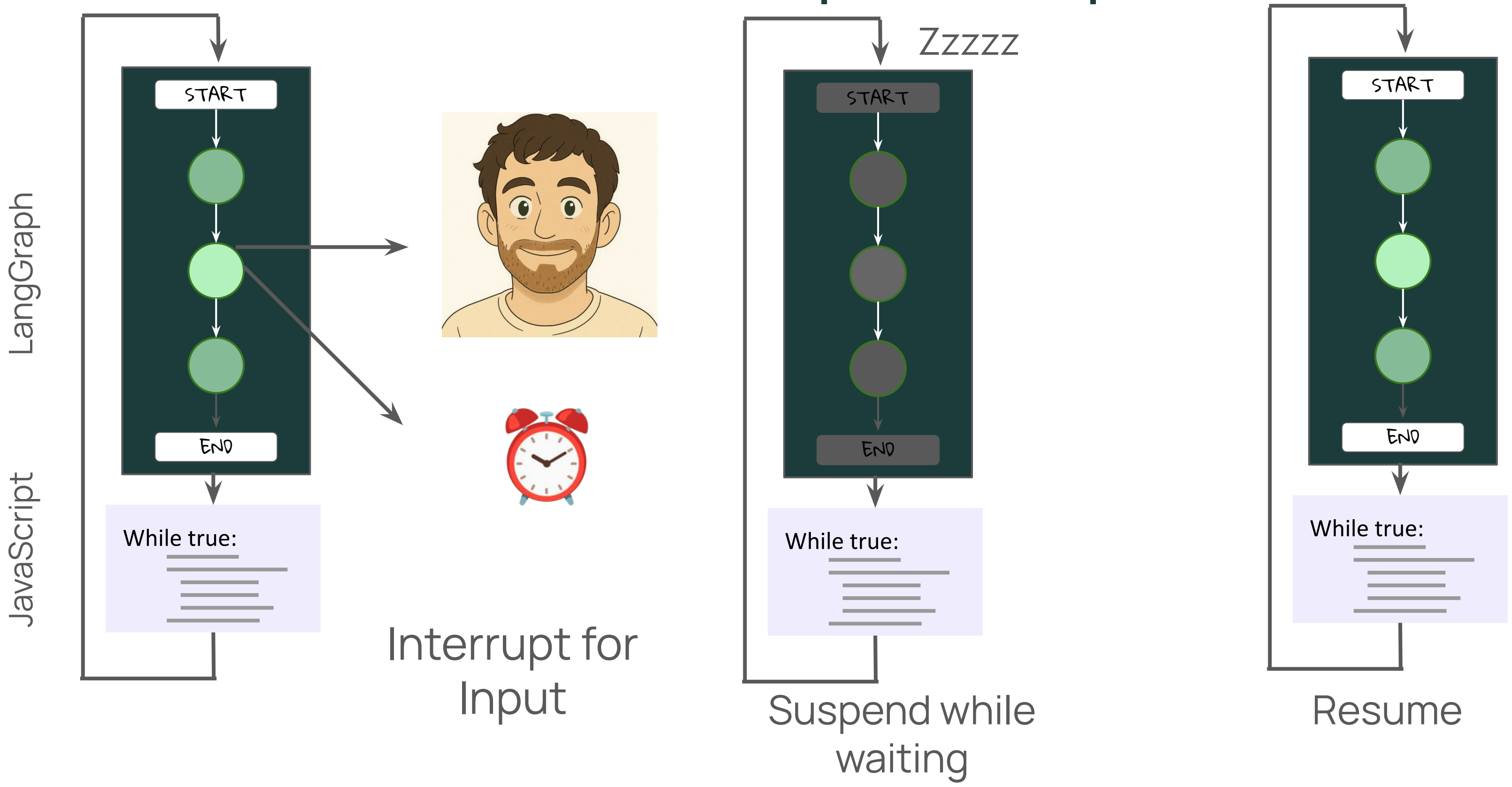
Benefits

- **Recover gracefully from failures** — resume without losing progress.
- **Time travel** — roll back to a known good point and continue forward.
- **Persistent state** — data is preserved even when the graph is not running.
- **Restore state at any step** — pick up execution from where you left off.



Interrupts

Human In the Loop: Interrupt



Application

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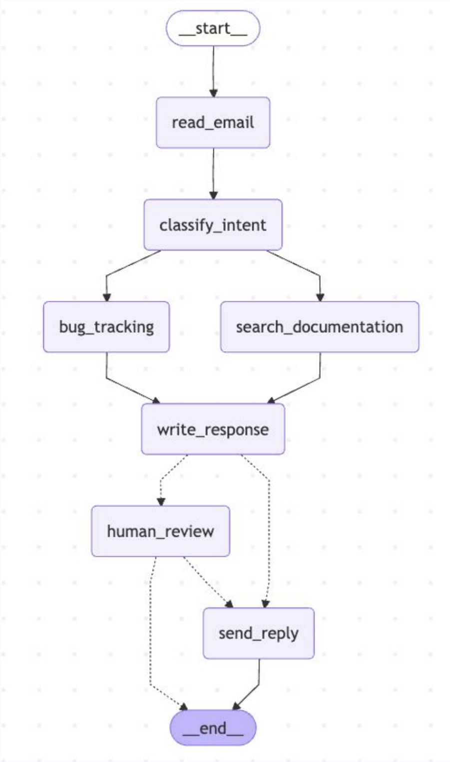
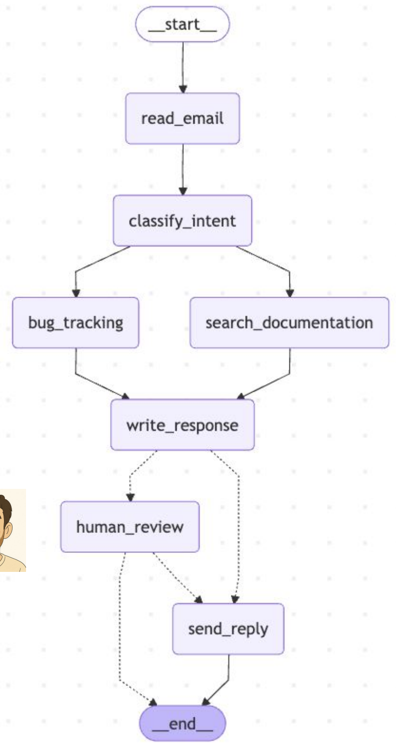


Email Support Workflow



Simulate a email customer support workflow

Focus on LangGraph aspects:

- State, Nodes,
- Edges
  - Serial
  - Parallel
  - Conditional
- Memory
- Interrupt

Conclusion

Congratulations!